

FedE-ator: A Reproducible Framework for Studying Energy Consumption in Federated Learning Systems

– PECS 2026, Pisa, Italy –

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August 25, 2026



Outline

- ➊ Introduction: Background & Motivation
- ➋ Related work
- ➌ FedE-ator Framework
- ➍ Experiment
- ➎ Conclusion & Future work

① Introduction: Background & Motivation

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What is Energy Consumption in Federated Learning?

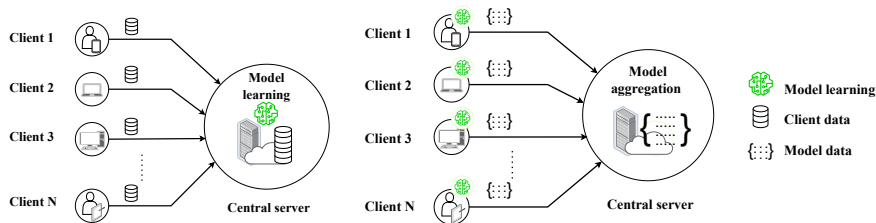


Figure 1: Left: Traditional Machine Learning: Data Comes to the Model → Data privacy issues. Right: Federated Learning framework: Model Goes to the Data → Keep data privacy.

Energy Consumption of FL $\neq$ Traditional ML

- Computation: distributes computation across many clients
- Communication: frequent lightweight model updates.

Energy in ML: the missing conversation

Figure 2. AI papers tend to target accuracy rather than efficiency. The figure shows the proportion of papers that target accuracy, efficiency, both or other from a random sample of 60 papers from top AI conferences.

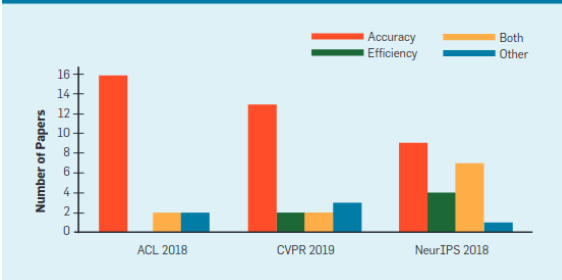


Figure 2: Only a small fraction of studies discuss both **energy & accuracy**¹.

- ML compute demands surged **300,000×** in recent years.
- Only a small fraction of studies discuss both **energy and model performance**.

¹Schwartz, Roy, et al. "Green AI." Communications of the ACM 63.12 (2020): 54-63.

Problem statement

The Problem

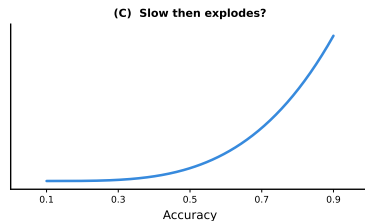
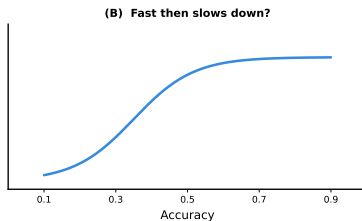
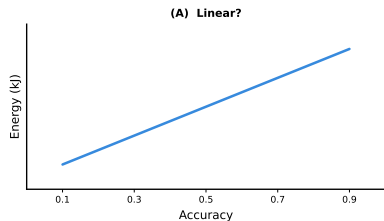
- Computational demands of state-of-the-art ML models have grown significantly
- Federated Learning (FL) distributes computation across many clients
- Energy cost = *computation* + *communication* across all nodes
- Only a small fraction of studies discuss the ML **accuracy-performance trade-off**

Key Point

Need a **framework to show the tradeoff** between **energy** and **FL performance** in **distribute environments**.

Energy - Accuracy tradeoff in FL?

What does the Energy-Accuracy curve look like?



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Related Work

Track A - energy measurement in FL

- Simulation only, no real measurement.
- For mobile phone only (e.g., CoconutBattery)
⇒ missing real distributed FL measurement

no fit ↓

Track B - tools found in FL experiments

- **HW-based:** External wattmeters, BMCs, PDUs
- captures full-device power
- **SW-based:** OS-level sensors (RAPL, NVML, /proc) - component-level insight
⇒ Measure in isolated devices, no sync

Remaining gaps

- ① Almost simulation only
- ② Applied to *isolated* devices - not full distributed FL, **Synchronization** of energy with training phases unsolved

FedE-ator addresses all these gaps with a unified, reproducible, synchronized HW+SW monitoring framework.

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FedE-ator: Overview & Architecture

Three-layer architecture:

- 1 **Layer 1 - FedE-ator Core:** handles input/output, orchestrates experiment campaigns
- 2 **Layer 2 - Monitoring Module - Expetator^a:** uses both SW and HW methods, synchronized via shared timestamps
- 3 **Layer 3 - FL Module:** SSH-based execution on server/client nodes

^a<https://expetator.readthedocs.io/>

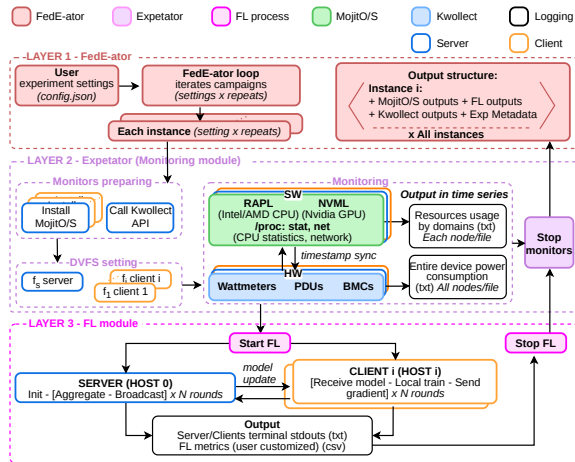


Figure 3: FedE-ator architecture

Framework Features

Framework functions

- **DVFS-based CPU/GPU heterogeneous frequency control:** adjusts frequencies per instance, restores defaults upon completion
- **Reproducibility:** JSON-driven configuration, multiple instances x repetitions in one campaign
- **Flexible FL configuration:** intra-framework (client scaling), inter-framework (adapts for differences across FL frameworks)

Framework limitations

- **Infrastructure requirements:** Linux-based distributed systems, supports SSH²+MPI³

²Secure Shell: protocol for remote login and command execution

³Message Passing Interface: standard for communication between processes in distributed/parallel computing

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Experimental Setup

Platform: Grid'5000/SLICES-FR⁴, a large-scale experimental testbed for parallel and distributed computing.

Chosen clusters:

- nova (CPU)
- gemini (GPU)
- gros (CPU)



⁴<https://www.grid5000.fr/w/Grid5000:Home>

Experimental Setup

Benchmark:

- **FL frameworks:** Flower⁵ & FedScale⁶
- **Model:** CustomCNN (LeNet5-based), MobileNetV2, ResNet18
- **Dataset:** CIFAR-10 (IID, fixed seed)
- **FL strategy:** FedAvg (all clients participate each round)

⁵<https://flower.ai/docs/framework/index.html>

⁶<https://github.com/SymbioticLab/FedScale>

Experimental Setup

Scenarios:

- **SC1 - Heterogeneous:** CPU+GPU nodes, 3 [CPU, GPU] freqs, 2 repetitions.
- **SC2 - Large-scale (stability testing):** 90 CPU nodes, $f_{\min}$ & $f_{\max}$.
- **SC3 - Cross-framework:** Flower vs. FedScale, 3 models, 4 freqs, 5 repetitions.

SC1 - Heterogeneous (for framework functionality validation)

Table 1: Settings

Server: nova-10
Clients: gemini-[1,2], nova-[11,12]
Freq. GHz: $f_{\min}$: [0.135, 1.2],
 f_{mid} : [0.832, 1.7], $f_{\max}$: [1.53, 2.1]

Framework: Flower
Strategy: FedAvg

Library: PyTorch
Model: CustomCNN
Local Epochs: 10
Rounds: 3

Repetitions: 2

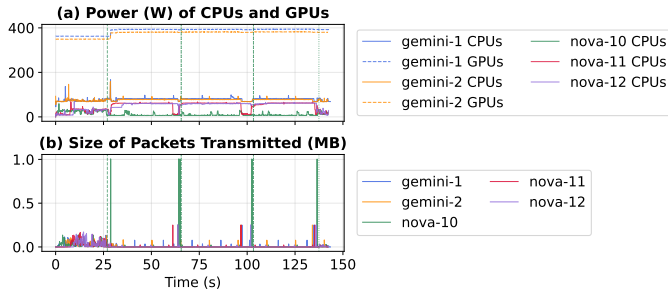


Figure 4: CPU/GPU Energy consumption and network traffic (SC1, $f_{\max}$).

Clients: **3 energy plateaus** matching 3 rounds, server: flatter profile. Network spikes align with model exchange. 10 Hz sampling captures fine-grained phase transitions.

SC1 - Heterogeneous (for framework functionality validation)

Table 2: Settings

Server: nova-10
Clients: gemini-[1,2], nova-[11,12]
Freq. GHz: $f_{\min}$: [0.135, 1.2], f_{mid} : [0.832, 1.7], $f_{\max}$: [1.53, 2.1]
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Rounds: 3
Repetitions: 2

Table 3: Results

Freq. [GPU, CPU] (GHz)	[0.135, 1.2]	[0.832, 1.7]	[1.53, 2.1]
Time monitored (s)	250	158	134
Time FL training (s)	231	145	112
Mean P-entire system (W)	350	359	377
Mean P [GPU] (W)	752.22	764.20	793.29
Mean P [CPU] (W)	82.46	89.44	99.74
Total E (ms. by SW) (kJ)	231	152	140
Total E (ms. by HW) (kJ)	442	286	254
CO ₂ (gCO ₂ eq)	3.8	2.4	2.1
Accuracy (3 rounds)	0.556	0.557	0.563

Higher frequency requires higher power (W) but less time processing $\rightarrow E = P \times T$

2× differences between HW and SW measurement methods (HW: +fan, mainboard, NIC, etc.)

SC2 - Large-scale (for stability testing)

Table 4: Settings

Server: gros-104
Clients: 89 nodes (gros)
Freq. GHz: $f_{\max}$: 2.2, $f_{\min}$: 1
Framework: Flower
Strategy: FedAvg
Library: TensorFlow
Model: CustomCNN
Local Epochs: 1
Rounds: 5
Repetitions: 1

Table 5: Results

CPU Frequency (GHz)	$f_{\min}$ (1)	$f_{\max}$ (2.2)
Time monitored (s)	502.30	231.78
Time FL processed (s)	468.59	220.11
Mean power (Kwollect) (W)	69.50	76.14
Total energy (RAPL) (kJ)	919.8	565.4
Total energy (Kwollect) (kJ)	3254.4	1630.8
CO ₂ (gCO ₂ eq)	28.02	14.04
Accuracy	0.10	0.10

Total energy at $f_{\min}$ nearly $2 \times f_{\max}$ (**idle power** across 89 nodes dominates longer runs).
 Framework stable at **90 nodes**.

SC3 - Cross framework

Table 6: Settings

Server: nova-14

Clients: nova-[15,16,17]

Freq. GHz: $f_{\min}=1.2$,
 $f_{\text{mid}}=\{1.5,1.8\}$, $f_{\max}=2.1$

Framework: Flower*, FedScale⁺
Strategy: FedAvg

Library: PyTorch

Model: CustomCNN, ResNet18,
MobileNetV2

Local Epochs: 1* / 512⁺

Rounds: 10

Repetitions: 5

Energy Consumption to Reach Target Accuracy under Different Configurations

Training Model · FL Frameworks · CPU Frequencies

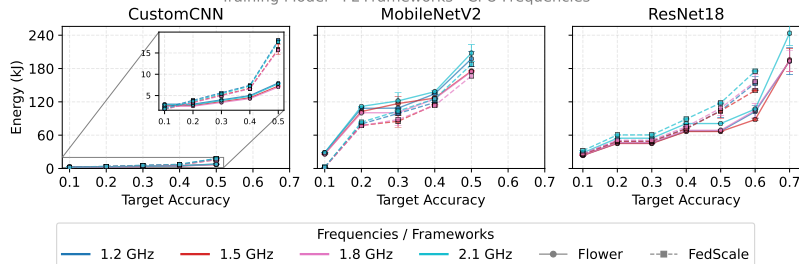
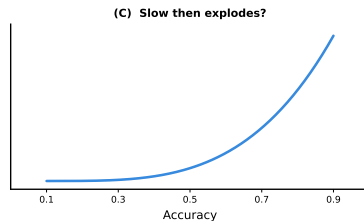
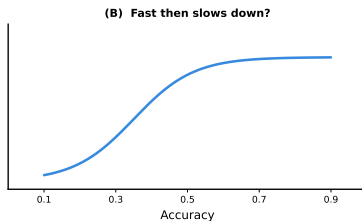
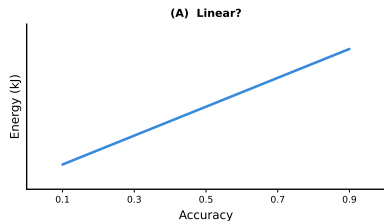


Figure 5: Energy vs. target accuracy across configs (SC3).

Energy remains relatively stable at low accuracy targets but increases sharply beyond a threshold $\rightarrow$ plan an accuracy target carefully.

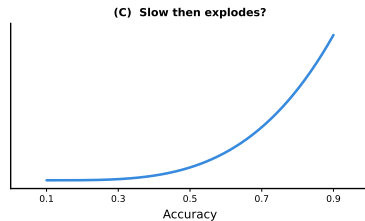
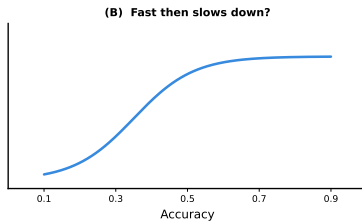
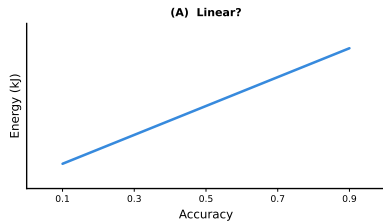
Energy - Accuracy tradeoff in FL?

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Energy - Accuracy tradeoff in FL?

What does the Energy-Accuracy curve look like?



C: Slow then explodes

SC3 - Cross framework: Energy vs Model complexity

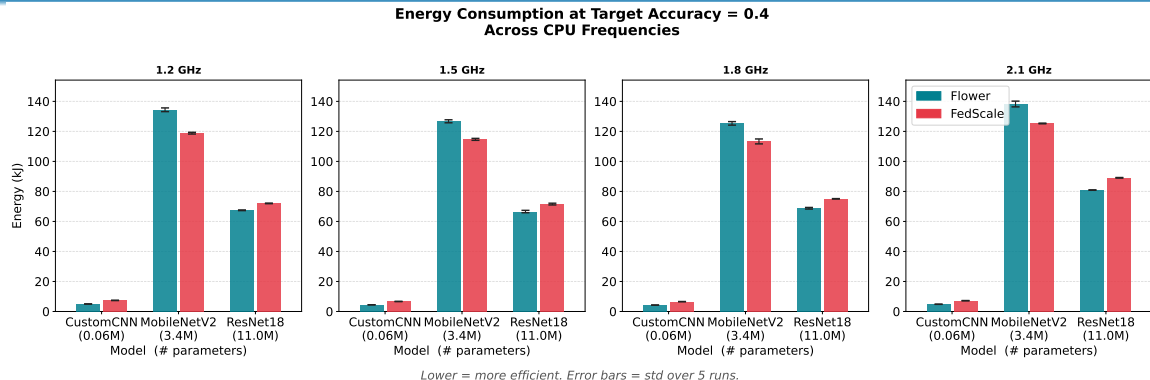


Figure 6: Energy vs ML Models (SC3).

MobileNetV2 is designed for mobile/inference (convolutions generate many small ops → inefficient on the server's GPU/CPU during training) → higher consumption.

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Conclusion & Future Work

Contributions

- ① **FedE-ator framework:** synchronized SW+HW energy monitoring, DVFS control, reproducible campaigns, adaptive FL configuration.
- ② **Experiments on Grid'5000** to validate the framework and obtain scientific insights.
- ③ **Insights:**
 - ▷ Optimal frequency $\neq$ max or min frequency (depends on models, tasks, HW, FL frameworks, *etc.*)
 - ▷ Marginal energy cost grows non-linearly with accuracy - the same accuracy increment is far cheaper at low accuracy than near convergence.

Future Work

- Upgrade FedE-ator with online DVFS-based energy efficiency for adaptive resource management (*i.e.*, realtime adjust frequencies base on system status)

Thank you for your attention.



This work was supported by the French National Research Agency (ANR) project Advancing Federated Learning while Reducing the Carbon Footprint - DELIGHT (ANR-22-CE23-0024).

Experiments were carried out using the Grid'5000 testbed, supported by a scientific interest group hosted by Inria and including CNRS, RENATER and several Universities as well as other organizations (see <https://www.grid5000.fr>).

